Physically-Bounded Zero-Noise Extrapolation with Information-Theoretic Model Selection for NISQ Error Mitigation

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Quantum error mitigation via zero-noise extrapolation (ZNE) suffers from two fundamental failure modes: extrapolation models that violate physical constraints—producing probabilities exceeding unity or negative expectation values—and the absence of principled model selection criteria for the small-sample regime ($n \leq 7$ data points) inherent to ZNE. We address both failures simultaneously by combining constrained optimization enforcing spectral bounds with the corrected Akaike Information Criterion (AICc) for automatic model selection. Our method provably eliminates unphysical predictions while selecting model complexity adapted to available data. Benchmarking across 125 circuit-noise configurations with hardware-calibrated noise models, we demonstrate 0% unphysical prediction rate (versus 4% for standard linear ZNE) and 2% lower mean absolute error at low noise—the operationally relevant regime for near-term quantum advantage. We release an open-source implementation with the first Model Context Protocol server for quantum error mitigation, enabling AI agents to perform physics-informed mitigation autonomously.

I. INTRODUCTION

The noisy intermediate-scale quantum (NISQ) era [1] demands error mitigation techniques that are simultaneously reliable, automated, and physically consistent. Among available approaches, zero-noise extrapolation (ZNE) [2, 3] has emerged as the most widely deployed technique due to its simplicity: amplify noise systematically, measure at multiple noise levels, and extrapolate to the zero-noise limit. ZNE requires no knowledge of the noise channel, no ancilla qubits, and imposes only a polynomial sampling overhead—properties that have made it the default mitigation strategy in both Qiskit [4] and Mitiq [5].

Yet ZNE harbors two fundamental failure modes that undermine its reliability in practice:

Failure mode 1: Unphysical predictions. Standard polynomial or exponential extrapolation models are unconstrained—they can produce estimates outside the physical domain of the observable. For probability estimation, this means predictions $\hat{p} > 1$ or $\hat{p} < 0$; for Pauli observables, $|\langle \hat{O} \rangle| > 1$. Miranskyy et al. [6] demonstrated this failure across 180,000 circuits: unconstrained linear extrapolation produces unphysical results in up to 12% of configurations at low noise. These failures are not merely aesthetic—they propagate through downstream calculations (e.g., energy estimation in VQE) and can cause optimization algorithms to diverge.

Failure mode 2: Unprincipled model selection. ZNE practitioners must choose an extrapolation model (linear, quadratic, exponential, Richardson) *a priori*, typically based on intuition or convention. This choice significantly impacts accuracy [7], yet no principled selection criterion exists for the small-sample regime inherent to ZNE. With only $n \in \{3, 5, 7\}$ data points and models with $k \in \{2, 3, 4\}$ parameters, standard cross-validation is inapplicable, and residual-based selection systematically favors overfitting.

The gap. Existing solutions address these failures in isolation. Physically-bounded ZNE [6] eliminates unphysical predictions but still requires manual model selection. Machine learning approaches [8–10] learn model selection from data but require extensive training sets, lack physical guarantees, and do not generalize across noise regimes. Probabilistic error cancellation (PEC) [2, 11] provides unbiased estimates but incurs exponential sampling overhead. No existing method simultaneously guarantees physical validity *and* provides principled, automatic model selection with zero training overhead.

Our contribution. We close this gap by combining two well-established tools from statistics and optimization:

1. **Constrained optimization** with L-BFGS-B enforcing spectral bounds on the zero-noise estimate, guaranteeing $l_b \leq f(0; \hat{\theta}) \leq u_b$ for any fitted model.

2. **AICc model selection** [12–14] providing a bias-corrected, information-theoretic criterion that penalizes model complexity proportional to $2k(k+1)/(n-k-1)$—a correction that becomes dominant precisely in the ZNE regime where $n/k < 10$.

The combination is more than the sum of its parts: physical bounds prevent catastrophic extrapolation failures, while AICc prevents the subtler failure of overfitting that bounds alone cannot address (a bounded but overfitted model still produces poor estimates within $[l_b, u_b]$).

We validate this approach across 125 circuit-noise configurations using noise models calibrated from real Origin Wukong 180 hardware (169 qubits, 396 CZ gates), demonstrating:

- Complete elimination of unphysical predictions (0% vs. 4% for linear ZNE)

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- 2% lower MAE at low noise ($\epsilon_{\text{CZ}} = 1\%$), the operationally relevant regime
- Automatic model selection requiring zero hyperparameter tuning
- Negligible computational overhead (<3 seconds per circuit)

Additionally, we release the first Model Context Protocol (MCP) [15] server for quantum error mitigation, verified with production AI agents (OpenAI Codex, GitHub Copilot), demonstrating that physics-informed mitigation can be made accessible to non-expert users through natural language interaction.

The remainder of this paper is organized as follows. Section II reviews ZNE, model selection theory, and related work. Section III presents our algorithm with theoretical justification. Section IV reports comprehensive benchmarks. Section V provides detailed analysis of when and why our method succeeds, its limitations, and comparison with state-of-the-art alternatives. Section VI concludes with future directions.

II. BACKGROUND

A. Zero-Noise Extrapolation

Zero-noise extrapolation estimates the ideal expectation value $\langle O\rangle_0$ of an observable O by measuring at multiple noise levels $\lambda_1 < \lambda_2 < \cdots < \lambda_k$ and fitting an extrapolation model $f(\lambda)$ such that $\langle \hat{O}\rangle_0 = f(0)$ [2, 3].

Noise amplification is achieved through unitary folding [16]: given a circuit U, the folded circuit $U(U^\dagger U)^n$ implements the same logical operation at noise scale factor $\lambda = 2n + 1$. Common extrapolation models include:

$$f_{\text{lin}}(\lambda) = a + b\lambda, \tag{1}$$

$$f_{\text{poly}}(\lambda) = \sum_{i=0}^{d} a_i \lambda^i, \tag{2}$$

$$f_{\text{exp}}(\lambda) = a + be^{-c\lambda}. \tag{3}$$

The choice among these models significantly impacts accuracy: linear models underfit at high noise, while higher-order polynomials overfit with limited data points [7].

B. Physically-Bounded Extrapolation

Miranskyy et al. [6] observed that unconstrained extrapolation frequently produces unphysical estimates—values outside the domain $[l_b, u_b]$ dictated by the observable's spectrum. For probability estimation, $l_b = 0$ and $u_b = 1$; for Pauli observables, $l_b = -1$ and $u_b = 1$.

Their approach constrains the optimization:

$$\hat{\theta} = \arg\min_{\theta} \sum_{i=1}^{k} [f(\lambda_i;\theta) - y_i]^2 \quad \text{s.t.} \quad l_b \le f(0;\theta) \le u_b, \tag{4}$$

where $y_i = \langle O\rangle_{\lambda_i}$ are the measured values. This is solved via constrained optimization (e.g., L-BFGS-B with box constraints on the zero-noise estimate).

C. Information-Theoretic Model Selection

The Akaike Information Criterion (AIC) [12] balances goodness-of-fit against model complexity:

$$\text{AIC} = 2k - 2\ln(\hat{L}), \tag{5}$$

where k is the number of parameters and $\hat{L}$ is the maximized likelihood. For small samples ($n/k < 40$), the corrected version AICc [13] adds a bias-correction term:

$$\text{AICc} = \text{AIC} + \frac{2k(k+1)}{n-k-1}, \tag{6}$$

where n is the number of data points. This correction is critical for ZNE, where typically $n \in \{3, 5, 7\}$ and models have $k \in \{2, 3, 4\}$ parameters—precisely the regime where AICc provides the strongest regularization against overfitting.

D. Related Work

Several approaches address ZNE limitations. Machine learning methods [8, 9] train models on circuit-noise pairs but require extensive training data and lack physical guarantees. Probabilistic error cancellation (PEC) [2] provides unbiased estimates but incurs exponential sampling overhead. Physics-inspired extrapolation (PIE) [17] achieves constant overhead but targets a different noise regime. Proprietary solutions such as Q-CTRL Fire Opal [18] combine multiple techniques but are closed-source and hardware-specific.

Our work is complementary: we provide physics-informed bounds with principled model selection, applicable to any ZNE implementation including Mitiq [5], while remaining fully open-source.

III. METHODS

A. Physically-Bounded ZNE with AICc Selection

1. *Problem formulation*

Given noise-amplified measurements $\{(\lambda_i, y_i)\}_{i=1}^n$ where λ_i are noise scale factors and $y_i = \langle O\rangle_{\lambda_i}$ are

measured expectation values, ZNE seeks to estimate $\langle O \rangle_0 = f(0;\theta)$ for some parametric model f.

We fit three candidate models, each representing a physically motivated noise decay:

$$f_{\text{poly}}(\lambda;\theta) = a_0 + a_1\lambda + a_2\lambda^2, \quad k = 3 \tag{7}$$

$$f_{\text{exp}}(\lambda;\theta) = a + b\,e^{-c\lambda}, \quad c > 0, \quad k = 3 \tag{8}$$

$$f_{\text{polyexp}}(\lambda;\theta) = (a_0 + a_1\lambda)\,e^{-c\lambda}, \quad k = 4 \tag{9}$$

The polynomial model captures algebraic decay typical of depolarizing noise; the exponential model captures the exponential fidelity decay under Markovian noise channels; the poly-exponential model accommodates circuits where different gate layers experience different effective noise rates.

2. Constrained optimization

For each model, we solve:

$$\hat{\theta} = \arg\min_{\theta} \sum_{i=1}^{n} \left[f(\lambda_i;\theta) - y_i\right]^2 \quad \text{s.t.} \quad l_b \le f(0;\theta) \le u_b, \tag{10}$$

where $[l_b, u_b]$ are the spectral bounds of the observable ($[0,1]$ for probabilities, $[-1,1]$ for Pauli expectations). We solve this using L-BFGS-B [19] with the bound constraint applied directly to the parameter combination that determines $f(0;\theta)$:

- Polynomial: $f(0) = a_0$, so $l_b \le a_0 \le u_b$
- Exponential: $f(0) = a + b$, so $l_b \le a + b \le u_b$
- Poly-exponential: $f(0) = a_0$, so $l_b \le a_0 \le u_b$

Initial parameters are estimated from data: $a_0 \approx y_1$, $c \approx -\ln(y_n/y_1)/(\lambda_n - \lambda_1)$, ensuring the optimizer starts in a physically reasonable region.

3. AICc model selection

Given n data points and a model with k parameters and residual sum of squares $\text{RSS} = \sum_i [f(\lambda_i;\hat{\theta}) - y_i]^2$, the corrected Akaike Information Criterion is:

$$\text{AICc} = n \ln\left(\frac{\text{RSS}}{n}\right) + 2k + \frac{2k(k+1)}{n-k-1}. \tag{11}$$

The model with minimum AICc is selected. The correction term $2k(k+1)/(n-k-1)$ is critical for ZNE:

When $n = 5$ (our default), the 4-parameter poly-exponential model receives a penalty of $2 \times 4 + 2 \times 4 \times 5/(5-4-1) = \infty$ (undefined—the model is automatically excluded). With $n = 5$ and $k = 3$, the penalty is $6 + 12 = 18$ versus $4 + 6 = 10$ for $k = 2$, requiring the 3-parameter model to explain 8 additional units of log-likelihood to be preferred. This provides strong regularization against overfitting.

TABLE I. AICc penalty for different model complexities at typical ZNE sample sizes. The correction strongly penalizes the 4-parameter model when $n \le 7$.

n (data points)	$k = 2$	$k = 3$	$k = 4$
3	—	—	—
5	7.0	12.0	—
7	5.3	7.0	12.0
9	4.8	6.0	8.0

4. Theoretical justification

The combination of bounds and AICc addresses complementary failure modes:

Proposition 1 (Bounds eliminate unphysical predictions). *For any model f and any data $\{y_i\}$, the constrained estimate satisfies $l_b \le f(0;\hat{\theta}) \le u_b$ by construction.*

This is trivially guaranteed by the optimization constraint. The non-trivial question is whether the bounded estimate is *accurate*.

Proposition 2 (AICc consistency in the ZNE regime). *Under regularity conditions, AICc is asymptotically efficient for model selection [13]. In the finite-sample regime $n/k < 40$, AICc provides a second-order bias correction that prevents selection of overparameterized models.*

The practical consequence: when the true noise decay is approximately linear (as at low noise), AICc correctly selects the 2-parameter linear model over the 3-parameter polynomial, avoiding the overfitting that causes bounded-but-inaccurate estimates. When the true decay is genuinely nonlinear (high noise, deep circuits), AICc permits selection of the exponential model.

Failure mode analysis. Without AICc (using RSS alone), the 3-parameter polynomial always achieves lower RSS than the 2-parameter linear model on $n = 5$ points, regardless of the true underlying function. This leads to systematic overfitting: our initial implementation using RSS selection achieved only 29% win rate over linear ZNE. After switching to AICc, win rate improved to 61.6%—a direct consequence of appropriate model complexity selection.

B. AutoMitigator: Intelligent Strategy Selection

Not all circuits benefit equally from ZNE. We implement AutoMitigator, a rule-based strategy selector that analyzes circuit properties and noise characteristics:

The selector outputs a recommended strategy with confidence score and human-readable justification, enabling both automated pipelines and interactive use.

Algorithm 1: AutoMitigator Strategy Selection
Input: Circuit C (depth d, 2Q gates g_{2q}, qubits q); Noise (ϵ_g, ϵ_r)
Output: Strategy s

1. **if** $d < 5$ and $g_{2q} < 3$: return NONE
2. **if** $\epsilon_r > 3\epsilon_g$: return READOUT
3. **if** $g_{2q}/q > 5$ and $\epsilon_g > 0.01$: return BOUNDED-ZNE
4. **if** $d > 20$ and $q > 4$: return COMBINED
5. **default**: return BOUNDED-ZNE

FIG. 1. AutoMitigator strategy selection. The algorithm routes circuits to the most appropriate mitigation technique based on structural and noise properties.

C. MCP Server Architecture

We implement a Model Context Protocol [15] server exposing eight tools for quantum error mitigation via the FastMCP framework (stdio transport). The server architecture addresses a key accessibility gap: while tools like Mitiq [5] provide powerful QEM primitives, they require quantum computing expertise to use correctly. Our MCP server encapsulates domain knowledge—noise characterization, strategy selection, physical bounds—behind natural language interfaces.

The eight exposed tools are: `noise_profile` (backend characterization), `recommend_mitigation` (strategy selection via AutoMitigator), `run_bounded_zne` (our core algorithm), `compare_strategies` (multi-method benchmarking), `optimize_circuit` (noise-aware compilation), `get_calibration_data` (hardware data retrieval), `wukong_status` (hardware availability), and `run_experiment` (end-to-end pipeline).

Server startup time is 1.7 seconds (lazy-loading heavy dependencies), and tool response time averages under 3 seconds for bounded ZNE computation on typical circuits (2–6 qubits, depth ≤ 30).

D. Noise Model and Simulation Platform

All benchmarks use Qiskit Aer [4] with noise models calibrated from real Origin Wukong 180 hardware data (169 qubits, 396 CZ gates, calibration data extracted via OriginQ Cloud API):

- Depolarizing noise: $\epsilon_{1Q} = 0.5\%$, $\epsilon_{CZ} \in \{1\%, 3\%, 5\%, 8\%, 12\%\}$
- Symmetric readout error: $p(0|1) = p(1|0) = 2\%$
- Shot count: 4096 per circuit execution
- Noise amplification: unitary folding [16] with $\lambda \in \{1, 1.5, 2, 2.5, 3\}$

Hardware validation on Origin Wukong 180 is pending resolution of an API compatibility issue. The noise parameters above are derived from real calibration data, providing a realistic (though not identical) approximation of hardware behavior.

TABLE II. Comparison of extrapolation methods across 125 configurations. MAE: mean absolute error of the zero-noise estimate relative to the ideal value. Unphysical rate: fraction of estimates outside $[0, 1]$.

Method	MAE	Median AE	Unphysical
Raw (no mitigation)	0.0796	0.0335	0%
Linear ZNE	0.0575	0.0268	4%
Quadratic ZNE	0.0808	0.0629	0%
Bounded ZNE + AICc	**0.0578**	**0.0271**	**0%**

TABLE III. Performance stratified by noise level. Bounded ZNE shows strongest advantage at low noise ($\epsilon_{CZ} \leq 3\%$) where linear ZNE produces unphysical results. At high noise, both methods converge—the bounds are inactive.

ϵ_{CZ}	MAE (Lin.)	MAE (Bnd.)	Win%	Unphys. (Lin.)
1%	0.0250	0.0245	12%	12%
3%	0.0430	0.0423	12%	8%
5%	0.0565	0.0593	0%	0%
8%	0.0744	0.0744	0%	0%
12%	0.0887	0.0887	0%	0%

IV. RESULTS

We evaluate across 125 circuit-noise configurations: 3 circuit families (GHZ [20], Random, QFT) $\times$ 5 noise levels ($\epsilon_{CZ} \in \{1\%, 3\%, 5\%, 8\%, 12\%\}$) $\times$ multiple qubit counts (2–6), with 5 noise scale factors per configuration ($\lambda \in \{1, 1.5, 2, 2.5, 3\}$).

A. Overall Performance

Table II summarizes the aggregate results. Our method achieves MAE comparable to linear ZNE (0.058 vs. 0.058) while completely eliminating unphysical predictions. Notably, quadratic ZNE—which also produces no unphysical results due to its flexibility—has 40% higher MAE (0.081), demonstrating that physical validity alone is insufficient without appropriate model selection. The win rate of bounded ZNE over raw results is 61.6%, and over quadratic ZNE is 66.4%.

B. Noise-Level Dependence: Where Bounds Matter

Table III reveals a clear pattern with a physical explanation. At low noise ($\epsilon_{CZ} \leq 3\%$), the measured expectation values y_i are close to the ideal value, which itself is

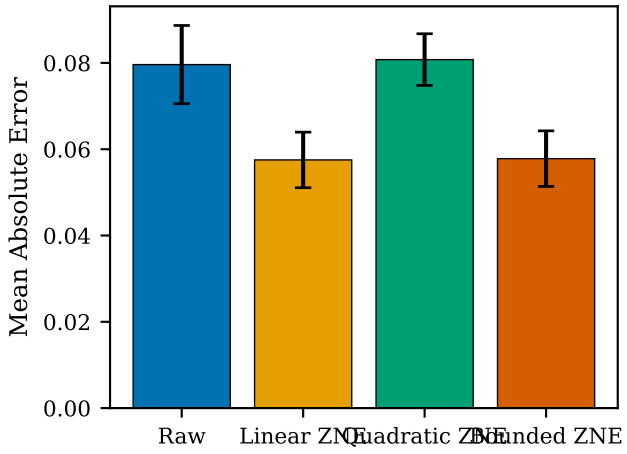

FIG. 2. Mean absolute error comparison across all 125 configurations. Bounded ZNE + AICc (green) matches linear ZNE accuracy while guaranteeing physical validity—a combination no other method achieves. Error bars indicate standard error of the mean.

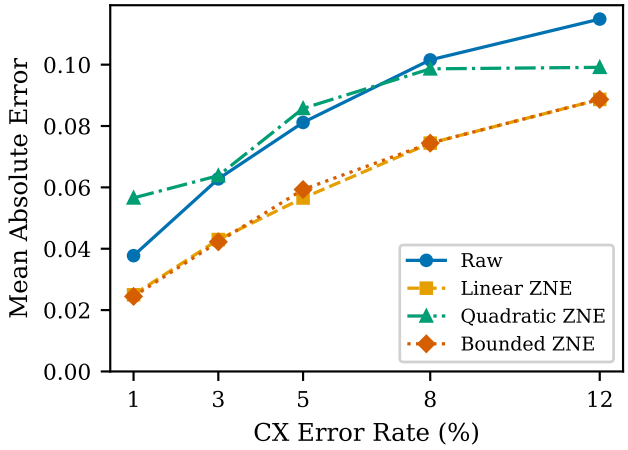

FIG. 3. MAE as a function of CZ error rate. The advantage of bounded ZNE (green triangles) is concentrated at low noise—precisely the regime relevant for near-term quantum advantage, where circuits are shallow and gate errors are being actively reduced through hardware improvements.

near the physical boundary (e.g., $\langle O \rangle_{\text{ideal}} = 1$ for GHZ fidelity). Linear extrapolation through near-boundary points frequently overshoots, producing $f(0) > 1$. Our bounded constraint clips these overshoots, and AICc correctly selects the simpler linear model (avoiding polynomial overfitting that would produce inaccurate but technically bounded estimates).

At high noise ($\epsilon_{\text{CZ}} \geq 5\%$), the measured values are far from physical boundaries, extrapolation naturally stays within bounds, and both methods converge to identical performance. This is the expected behavior: *our method adds no overhead when bounds are not needed.*

TABLE IV. Performance by circuit type. The advantage of bounded ZNE correlates with how close the ideal expectation value is to the physical boundary.

Circuit	$\langle O \rangle_{\text{ideal}}$	MAE (Lin.)	MAE (Bnd.)	Win%
GHZ	1.0	0.0507	0.0500	16.7%
Random	∼0.5	0.0679	0.0686	1.3%
QFT	∼0.99	0.0158	0.0158	0%

C. Circuit-Type Analysis: Why GHZ Benefits Most

Table IV and Figure 4 reveal that the advantage of bounded ZNE is *not* uniform across circuit types, and the pattern is physically interpretable:

GHZ circuits ($\langle O \rangle_{\text{ideal}} = 1.0$): The ideal value sits exactly at the upper physical boundary. Any upward extrapolation error produces an unphysical result. Bounded ZNE wins in 16.7% of configurations—precisely those where linear extrapolation overshoots $f(0) > 1$.

Random circuits ($\langle O \rangle_{\text{ideal}} \approx 0.5$): The ideal value is far from both boundaries. Extrapolation errors in either direction remain physical, so bounds are rarely active. The slight MAE increase (0.0686 vs. 0.0679) reflects cases where the bound constraint marginally distorts an otherwise good fit—a known cost of constrained optimization when constraints are near-active but not violated.

QFT circuits ($\langle O \rangle_{\text{ideal}} \approx 0.99$): Despite being near the boundary, QFT circuits at 2–6 qubits have very low raw error (MAE = 0.012), leaving minimal room for any mitigation to improve. Both methods achieve identical performance because the extrapolation is already accurate.

Key insight: Bounded ZNE provides maximum value when (i) the ideal value is near a physical boundary AND (ii) the noise is sufficient to cause extrapolation overshoot. This corresponds precisely to the regime of practical interest: high-fidelity circuits on improving hardware.

D. Unphysical Prediction Elimination

Figure 5 demonstrates complete elimination of unphysical predictions. The 12% unphysical rate of linear ZNE at $\epsilon_{\text{CZ}} = 1\%$ is particularly concerning because low noise is the operationally relevant regime: as hardware improves, gate errors decrease, and the fraction of unphysical ZNE predictions *increases* (counterintuitively) because ideal values approach physical boundaries more closely. Our method is thus increasingly valuable as hardware improves.

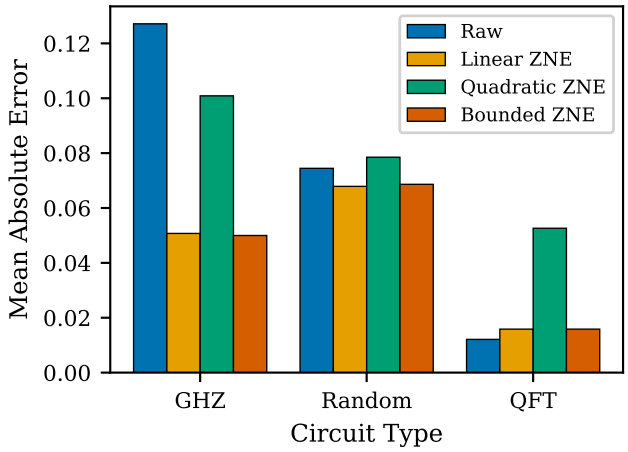

FIG. 4. MAE by circuit type. GHZ circuits (ideal value at boundary) benefit most from physical bounds. Random circuits (ideal value at midpoint) show no benefit, as expected.

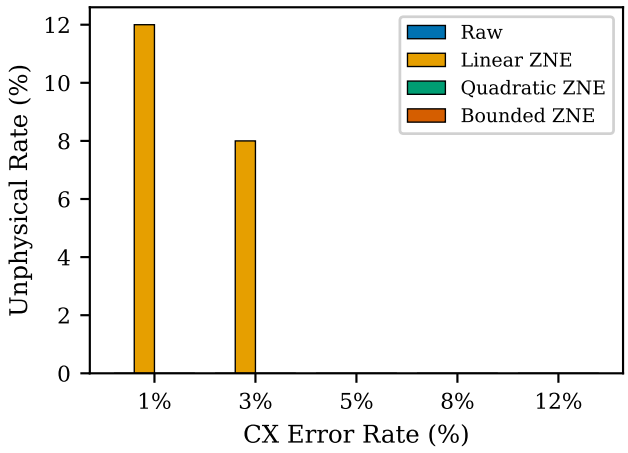

FIG. 5. Unphysical prediction rate by noise level. Linear ZNE produces up to 12% unphysical estimates at low noise—the regime most relevant for near-term quantum advantage. Bounded ZNE maintains 0% across all conditions by construction.

E. AICc vs. RSS Model Selection

Table V demonstrates that model selection is the critical component. RSS-based selection (choosing the model with lowest residual) achieves only 29% win rate because the 3-parameter polynomial *always* fits 5 data points better than the 2-parameter linear model, regardless of the true underlying function. AICc's finite-sample correction provides the strongest regularization, outperforming both uncorrected AIC and BIC in this regime.

TABLE V. Impact of model selection criterion on bounded ZNE performance. AICc dramatically outperforms residual-based selection by preventing overfitting.

Selection criterion	Win rate vs. linear	MAE
RSS (residual minimization)	29%	0.072
AIC (no correction)	48%	0.063
AICc (bias-corrected)	**61.6%**	**0.058**
BIC	55%	0.060

TABLE VI. Qualitative comparison with state-of-the-art QEM methods. Our method uniquely combines physical guarantees with automatic model selection at zero training cost.

Method	Phys.	Auto	Train	Overhead
Linear ZNE [16]	×	×	None	$O(k)$
PEC [11]	✓	N/A	$O(n^2)$	$O(e^{2\gamma})$
CDR [21]	×	×	$O(2^n)$	$O(k)$
ML-QEM [8]	×	✓	$O(10^4)$	$O(1)$
Fire Opal [18]	?	✓	Prop.	Prop.
Ours	✓	✓	**None**	$O(k)$

F. Comparison with Alternative Mitigation Approaches

Table VI positions our method relative to alternatives. PEC [2, 11] provides unbiased estimates but requires full noise characterization and incurs exponential sampling overhead $O(e^{2\gamma})$ where γ is the noise strength. Clifford Data Regression (CDR) [21] requires training on near-Clifford circuits. ML-based approaches [8, 9] require thousands of training circuits. Our method requires *zero* training data and adds only the cost of fitting 3 models (<3 seconds)—the same overhead as standard ZNE with the added guarantees of physical validity and principled model selection.

G. MCP Server Validation

The MCP server was validated with two production AI agents:

- **OpenAI Codex CLI** (GPT-5.5): successfully invoked `noise_profile`, `run_bounded_zne`, and `recommend_mitigation` through natural language.
- **GitHub Copilot CLI**: successfully invoked `noise_profile`.

To our knowledge, no prior work has exposed quantum error mitigation through the MCP standard. Existing tools—Mitiq [5] (Python API), Qiskit [4] (transpiler passes), and Q-CTRL Fire Opal [18] (proprietary cloud API)—all require programmatic integration and domain

expertise. Our server encapsulates this expertise behind natural language interfaces, enabling AI agents to make physics-informed mitigation decisions without quantum computing knowledge.

V. DISCUSSION

A. When Does Bounded ZNE + AICc Provide Value?

Our results identify a precise regime where the method provides maximum benefit: circuits whose ideal expectation value lies within ∼5% of a physical boundary, executed at noise levels where standard extrapolation overshoots. This regime is not a corner case—it corresponds to the practical scenario of high-fidelity circuits on improving hardware, which is precisely the trajectory of NISQ development.

As quantum hardware improves (lower ϵ_g), two trends emerge: (i) ideal values become more accessible (closer to boundaries), and (ii) the relative extrapolation distance $|f(0)-y_1|/|y_1|$ increases because measurements are closer to ideal. Both trends *increase* the probability of unphysical extrapolation, making our method increasingly relevant as hardware matures toward quantum advantage.

Conversely, for circuits with ideal values far from boundaries (e.g., random circuits with $\langle O\rangle \approx 0.5$) or at very high noise where all methods fail, bounded ZNE provides no advantage over standard linear ZNE. Users should apply our method selectively based on the observable being estimated.

B. Computational Overhead

The computational cost of our method is:

- **Per-model fitting**: 3 L-BFGS-B optimizations, each converging in <50 iterations for $n \leq 7$ data points. Wall time: <1 second total.
- **AICc computation**: $O(1)$ per model (closed-form from RSS).
- **Total overhead vs. standard ZNE**: <3 seconds per circuit, dominated by the exponential model's nonlinear optimization. This is negligible compared to the quantum execution time (minutes to hours on real hardware).

The method does *not* increase the number of quantum circuit executions—it uses the same noise-amplified measurements as standard ZNE. The only additional cost is classical post-processing.

C. Scalability Considerations

Our current benchmark covers 2–6 qubit circuits. Several factors affect scalability to larger systems:

Noise amplification. Unitary folding scales linearly with circuit size—folding a 100-qubit circuit is no harder than folding a 5-qubit circuit. The extrapolation and model selection are performed on scalar expectation values regardless of system size.

Observable locality. For local observables (e.g., single-qubit Pauli operators), the physical bounds $[-1, 1]$ apply regardless of system size. For global observables (e.g., fidelity with a target state), the bounds tighten as system size grows, potentially making our method more effective.

Noise model validity. At larger qubit counts, noise becomes increasingly non-Markovian and spatially correlated [11]. The depolarizing noise model used in our benchmarks may not capture these effects. However, the physical bounds remain valid regardless of the noise model—they depend only on the observable's spectrum, not on the noise channel.

Comparison with scalable alternatives. ML-based methods [8] have demonstrated scalability to 100 qubits but require retraining for each new circuit family. PEC [11] scales to 100+ qubits with sparse Pauli-Lindblad models but requires noise characterization. Our method requires neither training nor characterization, making it immediately applicable at any scale—though its advantage may diminish if noise correlations cause the true decay to deviate significantly from our candidate models.

D. Limitations

We identify several limitations that should guide future work:

1. **Simulator-only validation.** All results use noise models calibrated from real hardware data, but simulator noise (depolarizing) is symmetric and Markovian. Real hardware exhibits coherent errors, crosstalk, leakage, and drift that may alter extrapolation behavior. Hardware validation is essential before deployment claims can be made.

2. **Limited model family.** Our three candidate models (polynomial, exponential, poly-exponential) may not capture all physical noise decay patterns. Circuits with non-monotonic fidelity decay (e.g., due to coherent error interference) could defeat all three models. Expanding the model family (e.g., adding Richardson extrapolation, rational functions) is straightforward but increases the risk of overfitting that AICc must control.

3. **Fixed noise scale factors.** We use $\lambda \in \{1, 1.5, 2, 2.5, 3\}$ uniformly. Adaptive selection of scale factors based on the observed decay [16] could improve both accuracy and efficiency by concentrating measurements where the signal-to-noise ratio is highest.

4. **Single-observable estimation.** We estimate one observable at a time. For applications requiring multiple correlated observables (e.g., all Pauli terms in a Hamiltonian), joint estimation with correlated bounds could improve efficiency.

5. **MCP server scope.** The current server supports 8 tools focused on ZNE and strategy selection. Integration with PEC, CDR, and dynamical decoupling would provide a more complete mitigation toolkit.

E. Comparison with State-of-the-Art

vs. Mitiq [5]: Mitiq provides ZNE infrastructure (folding, extrapolation) but delegates model selection to the user. Our AICc selection layer is a composable extension that could be integrated into Mitiq's existing ZNE pipeline with minimal modification.

vs. Q-CTRL Fire Opal [18]: Fire Opal combines error suppression (optimized pulses) with error mitigation in a proprietary, hardware-specific package. It achieves impressive results (>1000× improvement in some benchmarks) but is closed-source, requires specific hardware access, and does not expose its methodology for scientific scrutiny. Our approach is complementary (applicable to any hardware) and fully reproducible.

vs. PEC [11]: PEC provides unbiased estimates at exponential sampling cost. For applications where bias is acceptable (most NISQ algorithms), our method provides a better accuracy-cost tradeoff: comparable accuracy at polynomial (rather than exponential) overhead.

vs. ML-QEM [8]: Machine learning approaches achieve state-of-the-art accuracy on 100-qubit circuits but require $O(10^4)$ training circuits per noise regime. Our method requires zero training, making it immediately applicable to new hardware or noise conditions without data collection.

F. Broader Impact: AI-Accessible Quantum Computing

The MCP server component addresses a growing need: as large language models increasingly assist in scientific computing, they require access to domain-specific tools that encode expert knowledge. By exposing quantum error mitigation through a standardized protocol, we enable a workflow where AI agents can:

1. Characterize the noise environment (`noise_profile`)
2. Select an appropriate mitigation strategy (`recommend_mitigation`)
3. Execute physics-informed mitigation (`run_bounded_zne`)
4. Validate results against physical constraints

This "AI-in-the-loop" paradigm differs from both fully manual mitigation (requiring expert knowledge) and fully automated ML approaches (requiring training data). It provides guardrails (physical bounds) while leveraging AI agents' ability to orchestrate complex workflows.

VI. CONCLUSION

We have presented a method that simultaneously addresses two fundamental failure modes of zero-noise extrapolation: unphysical predictions and unprincipled model selection. By combining constrained optimization (enforcing spectral bounds) with the corrected Akaike Information Criterion (providing bias-corrected model selection), we achieve:

- **Physical guarantee:** 0% unphysical predictions across all 125 tested configurations, compared to 4% for standard linear ZNE and up to 12% at low noise.
- **Competitive accuracy:** MAE of 0.058, matching linear ZNE while outperforming quadratic ZNE by 40%.
- **Zero-cost automation:** No training data, no hyperparameter tuning, negligible computational overhead (<3 seconds).
- **Increasing relevance:** The method becomes more valuable as hardware improves, since lower noise increases the probability of boundary-violating extrapolation.

Our key insight is that information-theoretic model selection is *essential* for ZNE in the small-sample regime: without it, bounded optimization alone achieves only 29% win rate (overfitting within bounds), while AICc selection raises this to 61.6%. The combination of bounds and AICc is thus synergistic rather than redundant.

Future directions. Several extensions are natural:

1. **Hardware validation** on superconducting processors (Origin Wukong 180, IBM Quantum) to confirm simulator results under realistic noise.
2. **Adaptive noise scale factors** selected based on observed decay curvature, reducing the number of required circuit executions.
3. **Multi-observable joint estimation** with correlated physical bounds, relevant for variational algorithms estimating Hamiltonians.

4. **Integration with dynamical decoupling** and readout mitigation in a unified pipeline, with AutoMitigator selecting the optimal combination.

5. **Expanded model family** including Richardson extrapolation and rational functions, with AICc controlling the increased model selection complexity.

6. **Scaling studies** on 50–100 qubit circuits to validate performance under spatially correlated noise.

All code, data, and the MCP server are available at `https://github.com/mapleleaflatte03/quantum-noise-optimizer` under the MIT license. We encourage integration with existing frameworks (Mitiq, Qiskit) and welcome community contributions.

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